AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

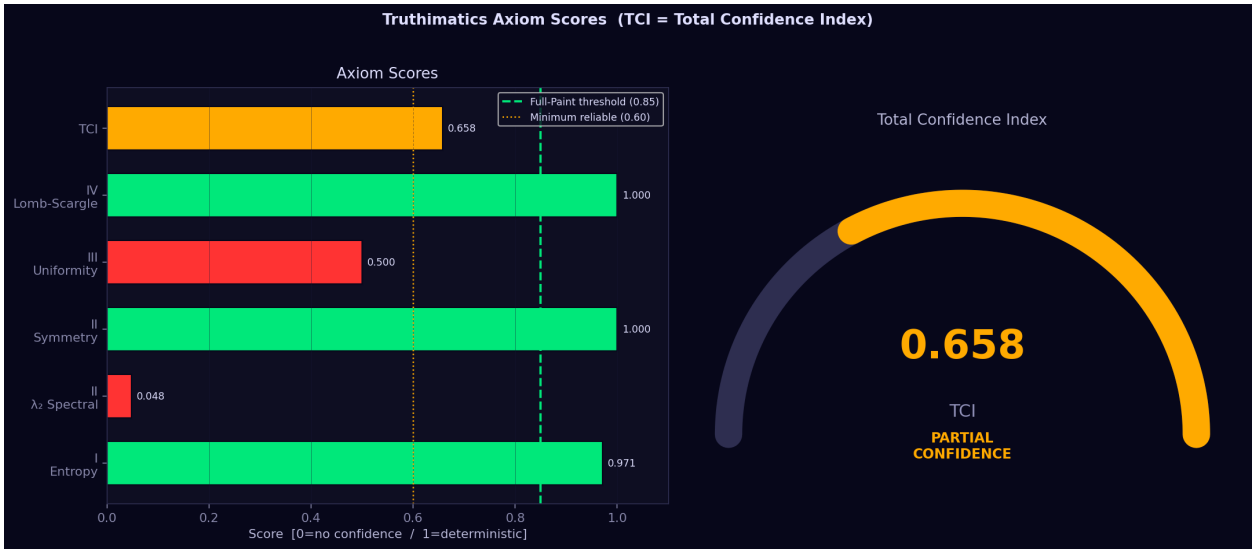
Specimen ID	AXIOM_EECA4E602F
Taxa	Archaeopteryx
Clade	Theropod
PBDB Records	10
Mass (regression)	0.5 kg (Campione & Evans 2012)
Mass (volumetric)	0.5 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	0.001 m³
Validity Score	1.0000 / 1.0000
TCI	0.6577 (partial)
Axiom I Entropy	0.9709
Axiom II λ■	0.1937 (symmetry 100%)
Axiom III Uniformity	0.5000
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	60.0% Measured / 40.0% Inferred
Elements	25
CoG (m)	[0.021, 0.048, 0.0]
Generated	2026-04-30 23:54 UTC

Ratio Violations

• metatarsal/tibia=0.724 outside [0.28,0.65]

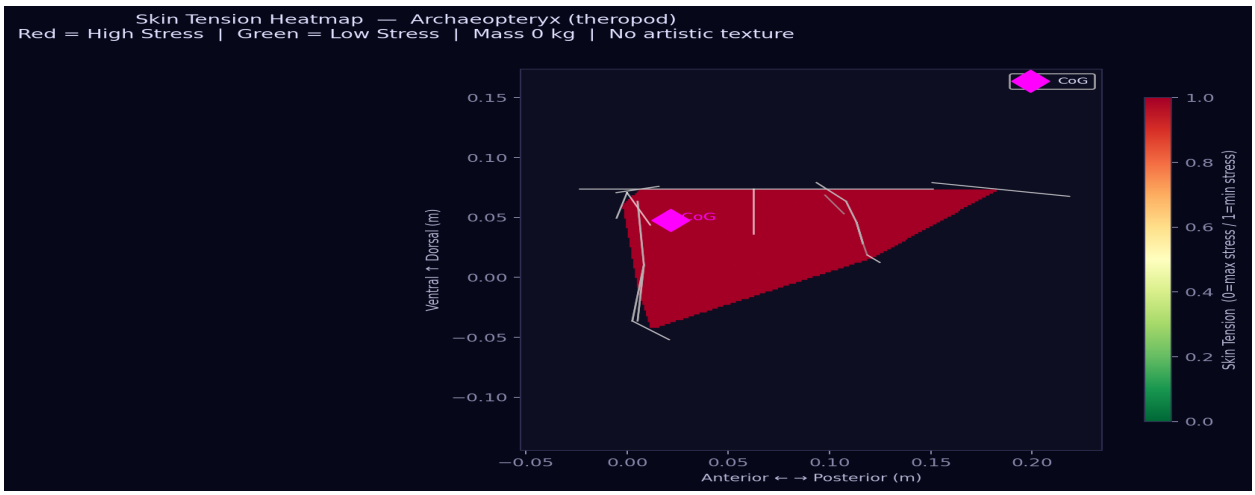
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



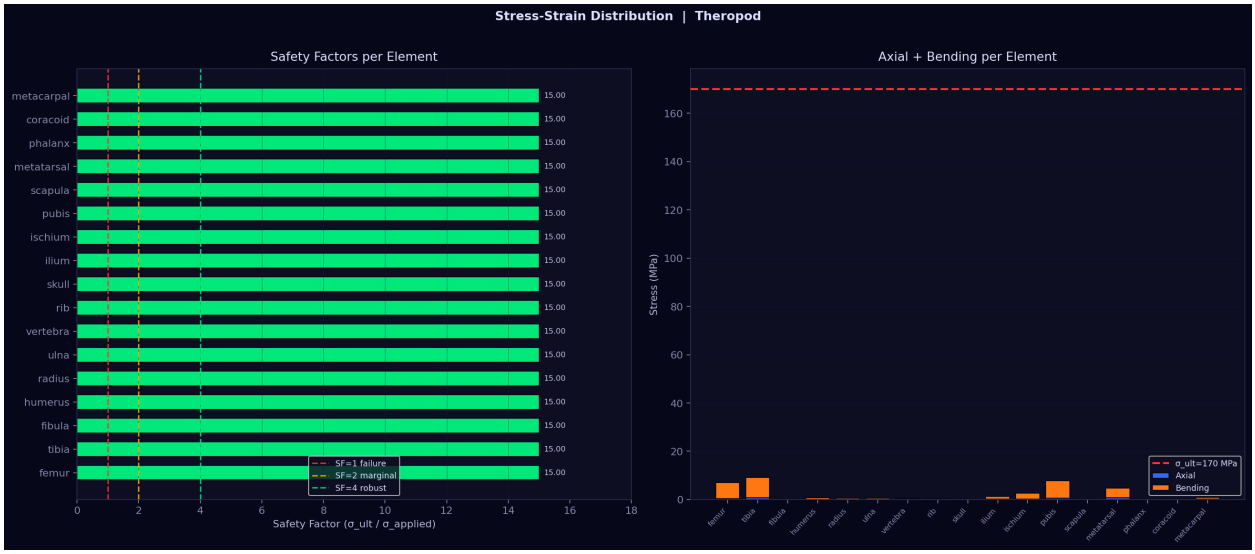
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

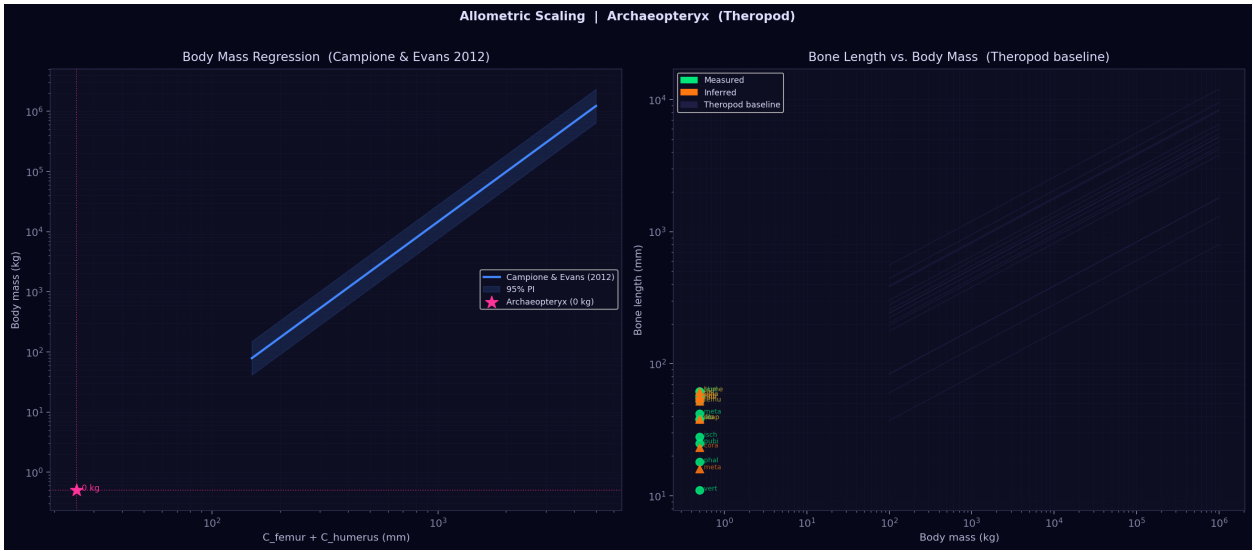


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.002	0.008	0.010	15.000	OK
femur	0.383	6.678	7.060	15.000	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.031	0.637	0.668	15.000	OK
ilium	0.170	1.157	1.327	15.000	OK
ischium	0.238	2.386	2.625	15.000	OK
metacarpal	0.099	0.806	0.905	15.000	OK
metatarsal	0.680	4.091	4.771	15.000	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	0.536	7.191	7.727	15.000	OK
radius	0.019	0.469	0.488	15.000	OK
rib	0.013	0.390	0.403	15.000	OK
scapula	0.002	0.009	0.010	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	0.680	8.474	9.154	15.000	OK
ulna	0.019	0.486	0.505	15.000	OK
vertebra	0.170	0.419	0.589	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.152 m	Femur + tibia + metatarsal
Estimated Speed	3.6 km/h (0.99 m/s)	Gait: Walk | Fr=0.661
Stride Length	0.308 m	Alexander (1976) / Thulborn (1990)
Bite Force	1.2 N (0.1 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	62.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2417	Circumference / length (dimensionless)
Limb Strength Index	1.6e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	0.53 m	Allometric from femur length
Est. Neck Length	0.07 m	Allometric from femur length
Est. Tail Length	0.2 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	23.2	8.9	0.73	0.9	Constraint	0.40	15.00
femur	52.0	4.0	0.33	0.4	Measured	0.88	15.00
femur_right	52.0	4.0	0.33	0.4	Symmetry	0.84	15.00
fibula	55.0	1.0	0.02	0.0	Measured	0.88	15.00
fibula_right	55.0	1.0	0.02	0.0	Symmetry	0.84	15.00
humerus	62.0	4.0	0.40	0.5	Measured	0.88	15.00
humerus_right	62.0	4.0	0.40	0.5	Symmetry	0.84	15.00
ilium	38.0	6.0	0.55	0.7	Measured	0.88	15.00
ilium_right	38.0	6.0	0.55	0.7	Symmetry	0.84	15.00
ischium	28.0	3.0	0.10	0.1	Measured	0.88	15.00
metacarpal	16.0	2.2	0.01	0.0	Constraint	0.65	15.00
metatarsal	42.0	3.0	0.15	0.2	Measured	0.88	15.00
phalanx	18.0	1.0	0.00	0.0	Measured	0.88	15.00
pubis	25.0	2.0	0.04	0.1	Measured	0.88	15.00
radius	55.0	3.0	0.20	0.2	Measured	0.88	15.00
radius_right	55.0	3.0	0.20	0.2	Symmetry	0.84	15.00
rib	38.0	2.0	0.06	0.1	Measured	0.88	15.00
scapula	38.0	10.0	1.52	1.9	Measured	0.88	15.00
scapula_right	38.0	10.0	1.52	1.9	Symmetry	0.84	15.00
skull	62.0	32.0	25.43	31.7	Measured	0.88	15.00
tibia	58.0	3.0	0.21	0.3	Measured	0.88	15.00
tibia_right	58.0	3.0	0.21	0.3	Symmetry	0.84	15.00
ulna	57.0	3.0	0.21	0.3	Measured	0.88	15.00
ulna_right	57.0	3.0	0.21	0.3	Symmetry	0.84	15.00
vertebra	11.0	6.0	0.16	0.2	Measured	0.88	15.00